

TURTLEGROVE

MAKING THINGS BEAUTIFUL TOGETHER: OPEN SOURCE, FROM THE VERY

P.A.M. VAN DAM

OCTOBER 9, 2026



Pascal
Van Dam

“Let us orchestrate your success!” #K8SMastery

INTRODUCTION

ABOUT THE TRAINER

- Pascal van Dam, living in Nieuw Bergen (Limburg/NL)
- Owner of Poortier Management B.V / PASCALVANDAM.COM
- Trainer, Architect & Consultant Open-Source Solutions:
 - Kubernetes & Containers
 - Virtualization & Cloud
 - Go, Rust, NodeJS, C, C++, Perl
 - Cloud Automation & Orchestration
 - CI/CD Argo, Flux, Gitlab
 - Linux Kernel Internals





Pascal Van Dam

“Let us orchestrate your success!” #K8SMastery

AGENDA

1. Who am I?
2. Why TurtleGrove?
3. Programming language archaeology
4. Introduction to the language
5. Logo as a serious language
5. Making it beautiful together
6. Demo
7. What's on offer
8. Questions

WHO AM I?

WHO AM I

- Learned programming at the age of 13
- Started with UCSD Pascal, Basic, Z80 Assembly
- Modula-2, C, C++, Python, Perl
- later: JavaScript, Go and Rust
- Officially a compiler builder
- Architect for Linux & K8s
- Trainer for K8s, Linux Kernel Internals, Go and Rust
- Proud father of 4 sons
- Love travelling, opensource and building with Legos
- Coaching gifted and HSP children

WHY TURTLEGROVE

IT STARTED AT THE KITCHEN TABLE

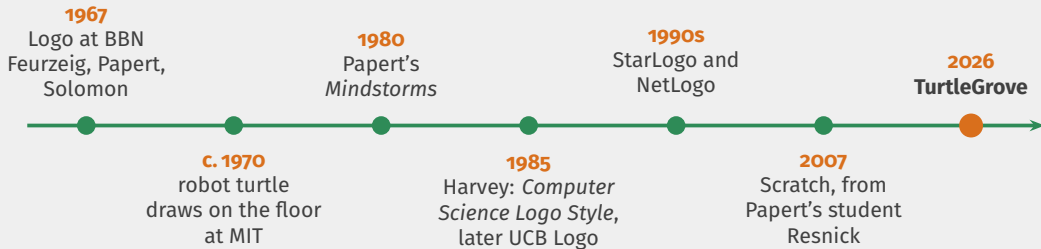
- A 9 year old kid asked me to learn him the art of programming
- Choices: logo, scratch, lego mindstorms etc.
- What I found: installers, accounts, English-only screens, and one child alone at one computer.
- Logo is legacy but still promising
- Can we make it align more to the philosophy behind it?
- Can we build it? Yes we can!
- The family became the test team: every release is tried at home

DESIGN CONSTRAINTS, IN ORDER

- **Must run on low resource commodity HW**
- **A URL is the whole deployment.** Zero install, works on a school Chromebook behind a school proxy.
- **The child's own language.** Six locales; errors are sentences, not stack traces.
- **The teacher sees the class.**
- **Children see *each other*.** This one changes everything.

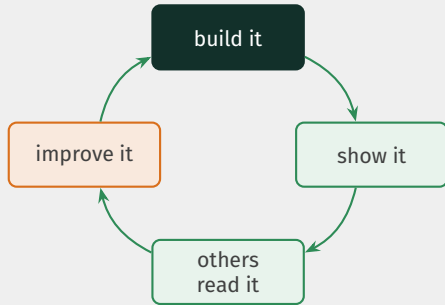
PROGRAMMING LANGUAGE ARCHAEOLOGY

ALMOST SIXTY YEARS OF TURTLES



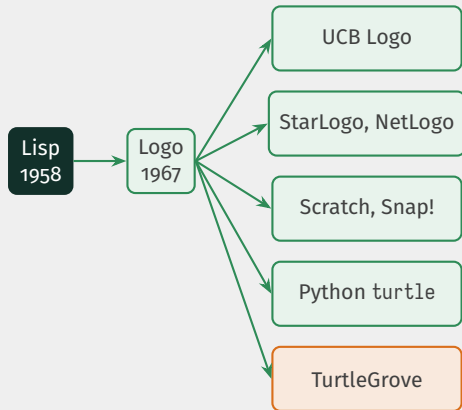
PAPERT'S BIG IDEA

- Papert had worked with **Jean Piaget**: children build knowledge, they do not receive it.
- **Constructionism**: learning happens best while *building* something you care about and can show to others.
- The child programs the computer, not the other way round.
- This loop is also the **open source loop**.



A LISP IN DISGUISE

- Logo is a dialect of **Lisp**, the language of early artificial intelligence.
- Words, lists, recursion, code as data. The parentheses were traded for a turtle.
- **Low floor:** `fd 100` is a program. **High ceiling:** recursion, higher-order procedures. **Wide walls:** whatever the child cares about.



INTRODUCTION TO THE LANGUAGE

FOUR WORDS MAKE A SQUARE

The turtle is always *somewhere*,
facing *somewhere*.

- fd walks, rt turns;
- stuck? *Play turtle*: walk the path yourself;
- repeat arrives the moment typing the same line gets boring.



square.logo

```
1  fd 100 rt 90
2  fd 100 rt 90
3  fd 100 rt 90
4  fd 100 rt 90
5
6  ; the same, once you are bored:
7  repeat 4 [fd 100 rt 90]
```

TEACHING THE TURTLE NEW WORDS

- to ... end defines a new word, exactly as real as fd;
- :n and :side are inputs;
- words build on words: a flower is squares that turn;
- now the experiments start: change 36 and 10 and see what grows.



flower.logo

```
1  to polygon :n :side
2    repeat :n [
3      fd :side
4      rt 360 / :n
5    ]
6  end
7
8  to flower
9    repeat 36 [ polygon 4 100 rt 10 ]
10 end
11
12 flower
```

LOGO AS A SERIOUS LANGUAGE

A TEN-YEAR-OLD WRITES RECURSION

There is no while in Logo. There does not need to be.



- tree draws a branch, then two smaller *trees*;
- if ... stop is the base case;
- the child *watches* the call stack go down a branch and come back.

tree.logo

```
1  to tree :size
2    if :size < 5 [ stop ]
3    fd :size
4    lt 30 tree :size * 0.7
5    rt 60 tree :size * 0.7
6    lt 30
7    bk :size
8  end
9
10 tree 90
```

LISTS, AND CODE AS DATA

TurtleGrove follows **UCB Logo**, the reference dialect.

- words, lists and arrays;
- map, filter, reduce, apply;
- a list of instructions is just a list:
you can run it.

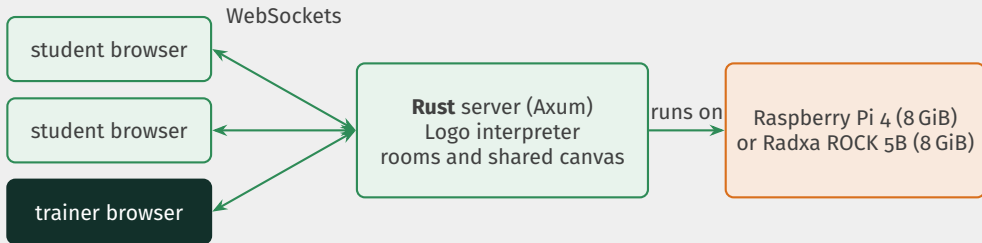
Harvey's *Computer Science Logo Style* still works.



lists.logo

```
1  to polygon :n :side
2    repeat :n [ fd :side rt 360 / :n ]
3  end
4
5  make "sides [3 4 5 6 8]
6  foreach :sides [
7    polygon ? 60
8    rt 360 / count :sides
9  ]
10
11 show map [? * ?] [1 2 3 4]
12
13 make "job [fd 50 rt 90]
14 repeat 4 [ run :job ]
```

SERIOUS ENGINEERING UNDERNEATH



- Front-end in plain **JavaScript**: no framework, no build step.
- The whole server side runs on a single **8 GiB** single-board computer: a Raspberry Pi 4 or a Radxa ROCK 5B is enough.
- **Open source**: a language and environment for learning should never be a black box.

BUILT LIKE PRODUCTION SOFTWARE

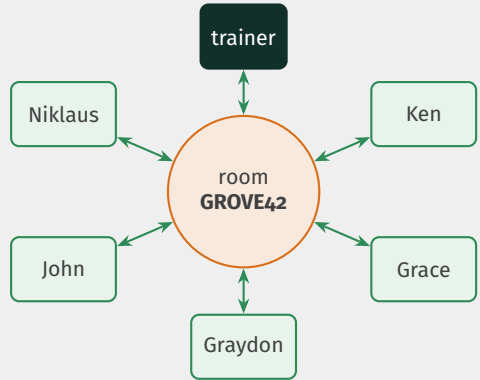


- GitLab CI on **arm64** and **amd64** Kubernetes runners, every stage pinned to a tool version.
- Images go to quay.io and a self-hosted registry, then roll out to Kubernetes.
- Children's code deserves the same supply-chain care as anyone else's.

MAKING IT BEAUTIFUL TOGETHER

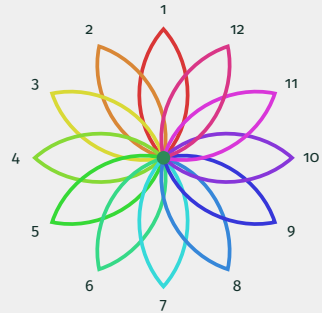
MANY TURTLES, ONE GROVE

- Join with a **name**, an **avatar** and a **room code**.
No account, no install.
- **Student** or **trainer**: the trainer guides, the children create.
- Every child's turtle is visible: you watch a classmate's spiral appear and want to know how.
- Papert's point exactly: children learn from each other's *artefacts*, not from each other's grades.



THE SHARED CANVAS

- One canvas, many turtles, drawing at the same time, in real time.
- Twelve children each drawing one petal makes a flower nobody could have drawn alone.
- Collaboration becomes a *property of the drawing*, not a classroom instruction.
- Also: excellent chaos. Which is also a lesson.



twelve children, one petal each

WHAT MAKES TURTLEGROVE DIFFERENT

Sharing code

- a link, and it runs on the other screen exactly as on yours;
- read it, run it, *change it*.

Asking for help

- A child can get inspiration from the examples
- A child can do the tutorials
- A child can ask help from the trainer
- stuck is normal; asking is part of the work.

Tutorials

- every step a runnable snippet: read a little, run a little, break it, fix it.

Examples and documentation

- every command, searchable, with an example you drop straight onto the canvas;
- written for a child, not for an implementer.

OPEN SOURCE, BEFORE THEY KNOW THE WORD

In the grove

share your code by choice

open a classmate's program

change it and make it yours

ask for help

say whose spiral you started from

In open source

publish your code

read someone else's code

fork and patch

open an issue

attribution and licences

Children learn the habits of open source long before they learn its vocabulary.

DEMO

LIVE DEMO: GROW A GARDEN TOGETHER

Open <https://turtlegrove.net>, join room **[ROOM CODE]**, and type along.

- everyone draws one polygon;
- everyone picks a colour from repcount;
- watch the grove fill up.

What happens if you change 30 to 31?



garden.logo

```
1  to polygon :n :side
2    repeat :n [ fd :side rt 360 / :n ]
3  end
4
5  to garden
6    repeat 12 [
7      setpc (hsv (repcount * 30) 100 100)
8      polygon 6 60
9      rt 30
10   ]
11 end
12
13 garden
```

WHAT'S ON OFFER

TEACHING CHILDREN IN AFRICA TO PROGRAM: OPEN SOURCE AFRICA

- A pan-African open source education community, based in **Lusaka, Zambia**.
- Trains locally through local instructors, works with universities to bring open source into the curriculum, and offers **free community training**.
- Grows local chapters that connect graduates to mentors and employers.
- TurtleGrove fits the brief: open, browser-based, free, and its server runs on one 8 GiB Raspberry Pi 4 or Radxa ROCK 5B.
- Next free online session: *Grooving in the TurtleGrove for Young Turtles*, 17 October 2026.

<https://opensourceafrica.org>

TAKE IT HOME, FOR FREE

- **The TurtleGrove environment:** open for any school, club or parent at <https://turtlegrove.net>.
- **Free course material:** PVD101 Programming in Logo, slides and exercises.
<http://https://github.com/pascal71/turtlegrove>
- **Free TurtleGrove sessions:** for schools, coding clubs and educators, online or on site.
Contact me @: <https://mastodon.social/@pamvdam>
- **Through Open Source Africa:** sessions for educators working in low-resource and low-bandwidth settings.

WHAT'S NEXT

1. **More translations:** for turtlegrove.net and for the courseware.
2. **A richer shared canvas:** enhance the collaboration experience.
3. **Closer to UCB Logo:** further improve compatibility.
4. **Train the trainer:** organise sessions so educators can run TurtleGrove themselves.
5. **More challenges:** starting with *Turtleballet*.

Want to help with any of these? Come and talk to me after the session.

QUESTIONS

Questions?

TurtleGrove	https://turtlegrove.net
Open Source Africa	https://opensourceafrica.org
Contact	https://mastodon.social/@pamvdam